


PGC306B: Tópicos Especiais em IA

Aprendizado de Máquina e Sistemas de Aprendizado Bio-inspirados

Intro to Machine Learning

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Objetivos

- Learn basic concepts about Machine Learning (ML);
- Discuss the past, present and future of ML;
- Show current ML applications;
- Show some well-reputed ML venues.

Roteiro

The Basics

Machine Learning: The Past

Machine Learning: The Present

Machine Learning: The Future

Well-reputed Venues

Overview about this course

Why Machine Learning?

- Recent progress in algorithms and theory
- Growing flood of online data
- Computational power is available
- AI boom (society, industry, research, etc.)

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- Recent progress in algorithms and theory
- Growing flood of online data
- Computational power is available
- AI boom (society, industry, research, etc.)
- **How about you?**

What is Machine Learning?

A multidisciplinary area

- Artificial intelligence
- Statistics
- Bayesian methods
- Computational complexity theory
- Control theory
- Information theory
- Philosophy
- Psychology and neurobiology
- ...

What is Machine Learning?

- Machine is easy . . . , but Learning?

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The Learning problem

- A computer program is said to learn from experience E with respect to some task T and some performance measure P if its performance on T, as measured by P, improves with E.

What is Machine Learning?

Learning to play checkers

- T?

What is Machine Learning?

Learning to play checkers

- T: play checkers
- P?

What is Machine Learning?

Learning to play checkers

- T: play checkers
- P: % of games won
- E?

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What is Machine Learning?

Learning to play checkers

- T: play checkers
- P: % of games won
- E: opportunity to play against self
  - What experience?
  - What exactly should be learned?
  - How shall it be represented?
  - What specific algorithm to learn it?

What is Machine Learning?

Learning to play checkers

- Type of training experience
  - Direct or indirect?
  - Teacher or not?

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- Choose the target function
  - ChooseMove: Board $\rightarrow$ Move

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Learning to play checkers

- Type of training experience
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- Choose the target function
  - ChooseMove: Board $\rightarrow$ Move
  - or V: Board $\rightarrow \mathcal{R}$
  - Analyzing a possible target function V:
    - if b is a final board state that is won, $V(b) = 100$
    - if b is a final board state that is lost, $V(b) = -100$
    - if b is a final board state that is drawn, $V(b) = 0$
    - if b is not a final board state, then $V(b) = V(b')$, where b' is the best final board state that can be achieved starting from b and playing optimally until the end of the game

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- Choose a representation for target function
  - collection of rules?

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  - neural network ?
  - polynomial function of board features?

Learning to play checkers

- A representation for learned function

$$w_0 + w_1 \cdot bp(b) + w_2 \cdot rp(b) + w_3 \cdot bk(b) + w_4 \cdot rk(b) + w_5 \cdot bt(b) + w_6 \cdot rt(b)$$

  - $bp(b)$: number of black pieces on board b
  - $rp(b)$: number of red pieces on b
  - $bk(b)$: number of black kings on b
  - $rk(b)$: number of red kings on b
  - $bt(b)$: number of red pieces threatened by black (i.e., which can be taken on black's next turn)
  - $rt(b)$: number of black pieces threatened by red

Learning to play checkers

- Obtaining Training Examples
  - $V(b)$: the true target function
  - $\hat{V}(b)$: the learned function
  - $V_{train}(b)$: the training value

What is Machine Learning?

Learning to play checkers

- Obtaining Training Examples
  - $V(b)$: the true target function
  - $\hat{V}(b)$: the learned function
  - $V_{train}(b)$: the training value
- One rule for estimating training values:
  - $V_{train}(b) \leftarrow \hat{V}(Successor(b))$

What is Machine Learning?

Learning to play checkers

- Least Mean Squares (LMS) weight update rule

Learning to play checkers

- Least Mean Squares (LMS) weight update rule
- Do repeatedly:
  - Select a training example b at random
    1. Compute $error(b)$:

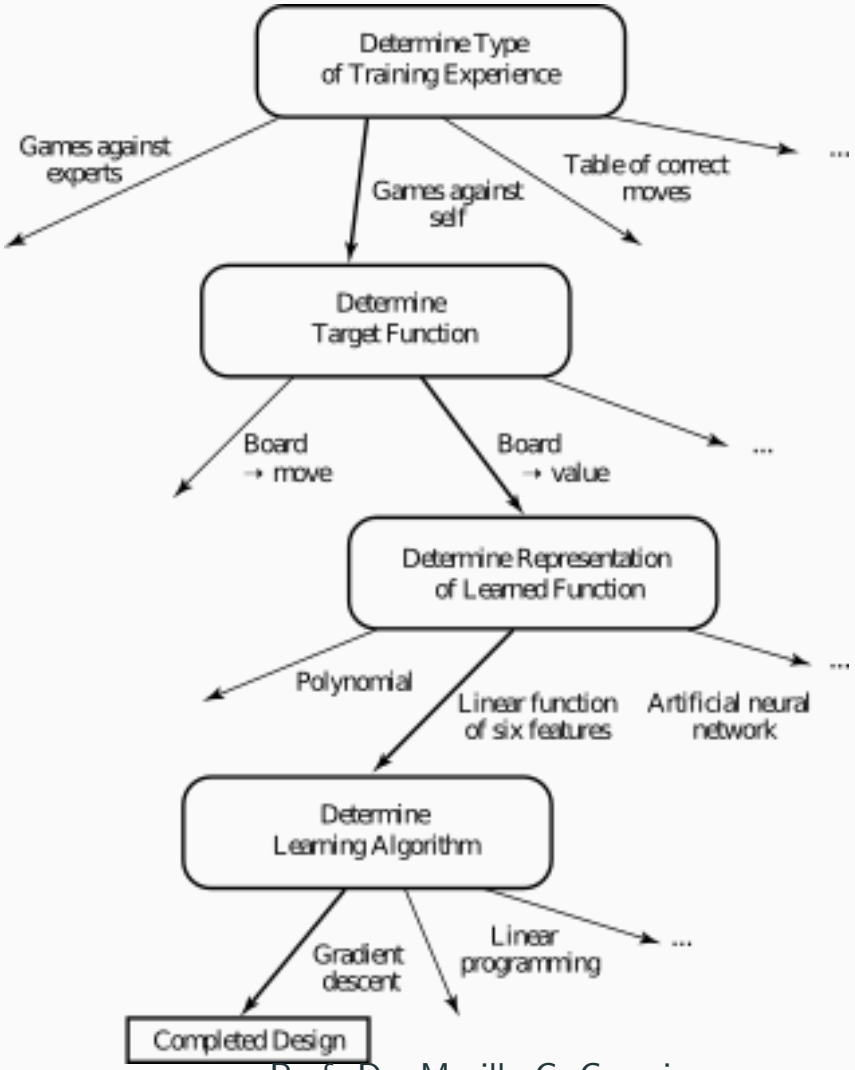
$$error(b) = V_{train}(b) - \hat{V}(b)$$

    2. For each board feature f_i, update weight w_i:

$$w_i \leftarrow w_i + c \cdot f_i \cdot error(b) \ ,$$

    with c being some small constant

Learning to play checkers - Summary of design choices



Roteiro

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- "Universal Machine" (Alan Turing, 1936)

ML History: Foundation

Neural Networks and AI

- The first neural network (Warren McCulloch and Walter Pitts, 1943)

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- Dartmouth College workshop (1956)

ML History: Enthusiasm and Expectations

Machine Learning and its promising results

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- Nearest neighbors (Thomas Cover and P. Hart, 1967)

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Falling down

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- Cut of investment for AI projects by the UK government (1973)

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- Backpropagation (David Rumelhart, Geoffrey Hinton and Ronald Williams, 1986)

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Great results!

- Commercial organizations meet Machine Learning

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- More resources available to Machine Learning research and development (datasets, competitions, etc.)

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Machine Learning everywhere

- Big data (2000s)

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- AlphaGo vs Lee Sedol (2016)

Roteiro

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Well-reputed Venues

Overview about this course

ML Today

Voice recognition

- Virtual assistant
- E.g., Siri, Google Now, Cortana ...

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  - "Where is the nearest gas station?"
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  - "Where is the nearest gas station?"
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- Assistant is able to find or relay phone information, or even send commands to other apps

ML Today

Self-driving cars

- Autonomous cars are getting closer and closer to become a reality

ML Today

Self-driving cars

- Autonomous cars are getting closer and closer to become a reality
- They learn to drive in the same way as humans, through experience
- Simulators and video games strongly assist in the accumulation of such an experience

ML Today

Games

- One of the most outstanding fields of application of AI
  - M. G. Carneiro. "Artificial intelligence in games evolution". Em: *Business, Technological, and Social Dimensions of Computer Games: Multidisciplinary Developments*. IGI Global, 2011, pp. 98–114
- Characters that analyze user behavior, respond to stimuli, react unpredictably, etc etc

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  - The system used a large search tree and examined 200 million moves per second

ML Today: Games

Games

- AlphaGo $\times$ Lee Sedol (Go)
  - Go is a complex board game considered harder than chess

ML Today: Games

Games

- AlphaGo $\times$ Lee Sedol (Go)
  - Go is a complex board game considered harder than chess
  - AlphaGo is the computer program developed by Google DeepMind to play Go
  - AlphaGo uses a combination of machine learning and tree searching techniques, combined through extensive training against humans and machines.

ML Today

Recommender systems

- Online businesses try to anticipate your need to buy a product so that you don't have to go to a store to buy it

ML Today

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ML Today

Recommender systems

- Online businesses try to anticipate your need to buy a product so that you don't have to go to a store to buy it
- Physical companies try to practice the same idea using discount coupons, so you can return more often
- There are a variety of ways to attract the customer: ads, discounts, etc.
- On the other hand, privacy is becoming increasingly rare. . .

ML Today

Recommender systems

- Netflix, spotify, pandora, ...
- Recommendation of a movie based on its history and past ratings

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ML Today

Many many other examples

- Autonomous planning and scheduling
- Spam detection, fraud
- Disease Diagnosis
- Robotics
- Machine translation
- *Smart homes*
- ...

Roteiro

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Well-reputed Venues

Overview about this course

ML Future

- Taking problems even more complex
- More industry solutions based on ML
- ML in questions regarding to privacy and security
- Efficiently learning from few data ("few shot learning")
- Better generalization ability for computer vision methods
- Better understandment about the human brain
- Do what humans do in a broader way (Geoffrey Hinton)
- ...

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ML Top Conferences and Events

- International Conference on Machine Learning (ICML)
- European Conference on Machine Learning and Knowledge Discovery in Databases (ECML-PKDD)
- International Joint Conference on Neural Networks (IJCNN)
- International Conference on Learning Representations (ICLR)
- ACM Conf. on Knowledge Discovery and Data Mining (KDD)

ML related

- AAAI Conference on Artificial Intelligence (AAAI)
- Conf. on Computer Vision and Pattern Recognition (CVPR)
- International Joint Conference on AI (IJCAI)
- Brazilian Conference on Intelligent Systems (BRACIS)*

ML Top Journals

- IEEE Trans. on Neural Networks and Learning Systems
- IEEE Trans. on Pattern Analysis and Machine Intelligence
- The Journal of Machine Learning Research
- Knowledge-Based Systems
- Neural Networks

ML related

- IEEE Trans. on Knowledge and Data Engineering
- IEEE Trans. on Cybernetics
- Expert Systems with Applications
- Neurocomputing

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See the course program available at moodle.

Referências

- T. M. Mitchell. *Machine Learning*. McGraw-Hill, 1997
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- A history of machine learning, Google, available at `https://cloud.withgoogle.com/build/data-analytics/explore-history-machine-learning/`
- Timeline of machine learning, Wikipedia, available at `https://en.wikipedia.org/wiki/Timeline_of_machine_learning`